

Decoder-only vs Encoder-Decoder with realtime example

This is one of the most common interview questions in Generative AI, LLM, and NLP interviews.

The easiest way to remember it is:

- **Decoder-only = Generate text**
- **Encoder-Decoder = Understand first, then generate**

High-Level Comparison

Feature	Decoder-Only	Encoder-Decoder
Purpose	Text Generation	Input → Output Transformation
Reads Input	Yes (as prompt)	Encoder reads completely
Generates Output	Yes	Yes
Best For	Chatbots, Code Generation	Translation, Summarization
Examples	GPT, Llama, Mistral, DeepSeek	T5, BART, FLAN-T5

Real-Time Example 1: ChatGPT

User asks

Explain Docker in simple words.

Decoder-Only

Prompt

Explain Docker in simple words.

The model predicts

Docker is a platform that packages applications
inside lightweight containers...



Here,

It simply keeps predicting the next word.

Explain

↓

Docker

↓

is

↓

a

↓

platform

↓

that

↓

packages

↓

...



There is **no separate encoder**.

Everything is treated as one long sequence.

Real-Time Example 2: Language Translation

User says

English

I love machine learning.



Expected output

French



J'aime l'apprentissage automatique.

Encoder-Decoder Flow

Input



I love machine learning



Encoder

(understands meaning)



Context Vector



Decoder

(generates French)



J'aime l'apprentissage automatique.

Notice

Encoder first understands.

Decoder then generates.

Real-Time Example 3: Email Summarization

Input

Dear team,



Yesterday we discussed the deployment plan...

...

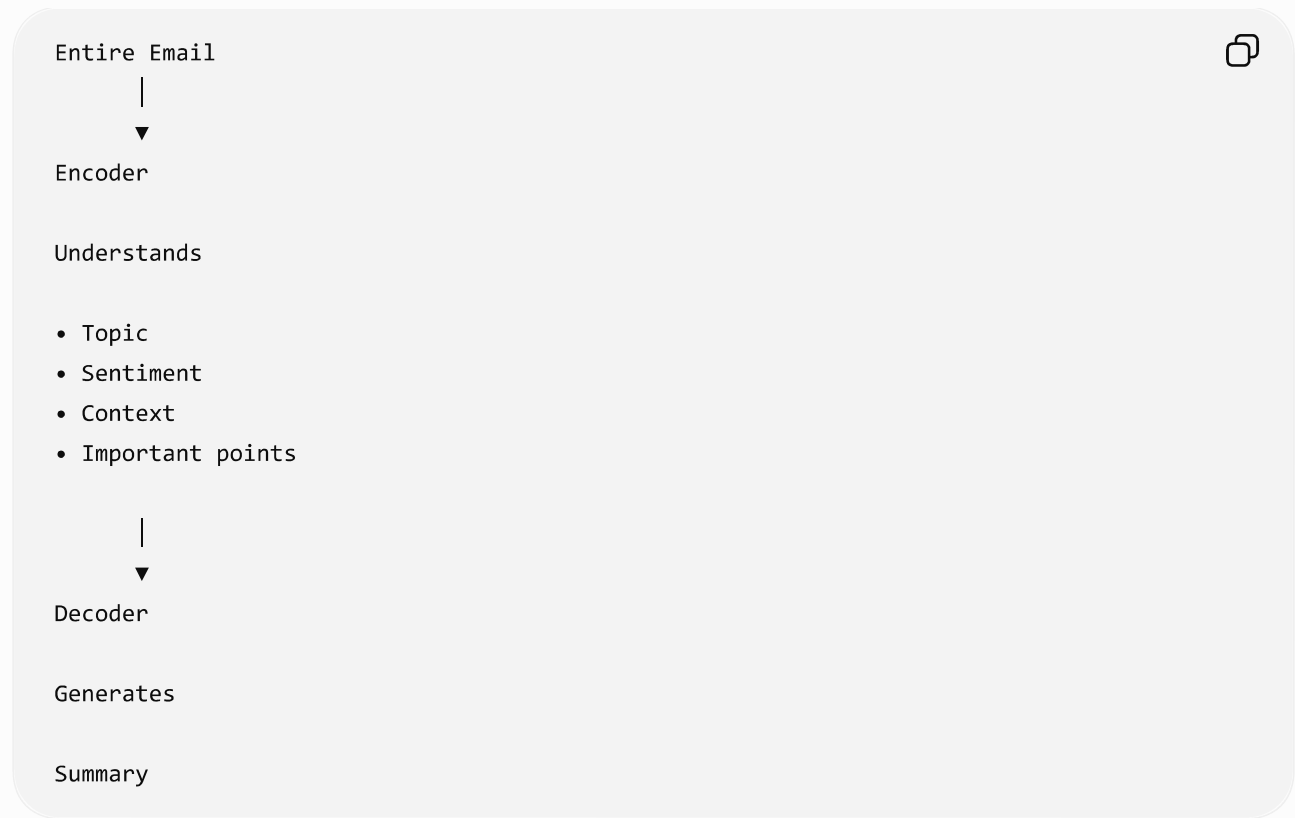
Output

Meeting summary:



Deployment starts Monday.
QA approval pending.

Process



Again

Encoder understands.

Decoder writes.

Real-Time Example 4: ChatGPT Conversation

You ask

Write Angular Signals example.

Prompt becomes

System Prompt

+

Previous Conversation

+

Current Question

The decoder receives everything.

System Prompt

↓

History

↓

Current Prompt

↓

Generate next token

↓

Generate next token

↓

Generate next token



No encoder.

Only prediction.

Decoder-Only Architecture

Input Prompt

What is Docker?

|



Tokenizer

↓

[What]

[is]

[Docker]

[?]

↓

Embedding



↓

Transformer Decoder

↓

Predict Next Token

↓

"is"

↓

Predict Again

↓

"a"

↓

Predict Again

↓

"container"

↓

Predict Again

↓

...

Everything is generated token by token.

Encoder-Decoder Architecture

Input



Translate to French:

I love AI.

|
▼
Encoder

Reads ALL tokens

↓

Creates Context Representation

↓

Decoder Starts

<START>

↓

J'

↓

aime

↓

l'

↓

IA

↓

<END>

Encoder processes the whole input before decoding begins.

Attention Difference

Decoder-Only

Uses

Masked Self-Attention



Meaning

Current token can only see previous tokens.

Example

I
love
machine
learning



Token "learning"

Can see

I
love
machine

Cannot see

future words

This prevents "looking ahead" during generation.

Encoder-Decoder

Encoder

Self Attention



Every word sees every word.

Decoder

Uses two attention mechanisms:

1. Masked Self-Attention

Generated words only



2. Cross Attention

Looks at encoder output.

Decoder



↓

Cross Attention

↓

Encoder Output

↓

Input Sentence

That's how translation works.

Real-Life Analogy

Decoder-Only

Imagine writing a story.

Once upon a time...



↓

write next sentence

↓

write next sentence

↓

write next sentence

You only use what has already been written.

Encoder-Decoder

Imagine translating a document.

First

Read the entire document.

Then

Write the translated version.

Exactly what humans do.

Popular Models

Decoder-Only

- GPT-3
- GPT-4
- GPT-5 family
- Llama 2/3
- Mistral
- Mixtral
- DeepSeek
- Qwen
- Gemma

Use cases:

- Chatbots
 - Coding assistants
 - Story generation
 - AI agents
 - Question answering
-

Encoder-Decoder

- T5
- FLAN-T5
- BART
- mBART
- PEGASUS

Use cases:

- Translation
- Summarization

- Grammar correction
- Question generation
- Text transformation

Which One Should You Use?

Task	Best Architecture	Why
Chatbot (ChatGPT)	Decoder-Only	Autoregressive text generation
Code Generation (GitHub Copilot)	Decoder-Only	Predicts the next token of code
Story Writing	Decoder-Only	Continuous text generation
Email Reply	Decoder-Only	Generates natural language responses
Translation	Encoder-Decoder	Understands source text before generating target text
Summarization	Encoder-Decoder	Reads the full document, then condenses it
Grammar Correction	Encoder-Decoder	Transforms one text into another
Paraphrasing	Encoder-Decoder	Rewrites while preserving meaning

Interview Answer (60 seconds)

Decoder-only models are designed for autoregressive generation. They take the input prompt and generate text one token at a time using **masked self-attention**, so each token can only attend to previously generated tokens. This architecture is ideal for chatbots, code generation, and creative writing. Examples include GPT, Llama, Mistral, DeepSeek, and Qwen.

Encoder-decoder models first use an encoder to read and understand the entire input sequence. The decoder then generates the output using both **masked self-attention** and **cross-attention** over the encoder's representations. This makes them well suited for sequence-to-sequence tasks such as translation, summarization, and grammar correction. Examples include T5, FLAN-T5, and BART.

A simple way to remember the difference is: **decoder-only models generate from a prompt, whereas encoder-decoder models first understand the input and then generate the output.**



give me a real life example to differentiate Decoder-only models vs Encoder-decoder models

The easiest way to differentiate them is through everyday scenarios rather than AI terminology.

Example 1: Customer Support (Best Interview Example)

Decoder-Only (Like ChatGPT)

You ask:

"My laptop is running very slowly. What should I do?"

The AI sees your question and immediately starts generating the answer word by word.

User Question



Decoder



Try restarting your laptop...



It predicts the next word based on everything it has seen so far.

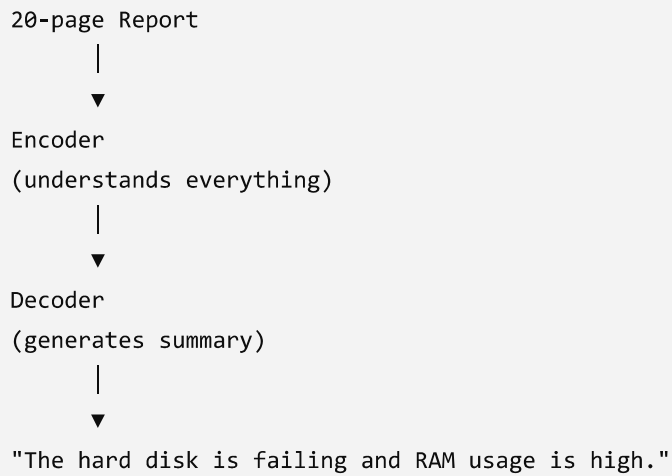
Think of it as: someone listening to you and replying naturally in a conversation.

Encoder-Decoder

Now imagine you upload a **20-page laptop diagnostic report** and ask:

"Summarize the issue."

The model first reads the **entire report**.



It doesn't start generating until it understands the whole document.

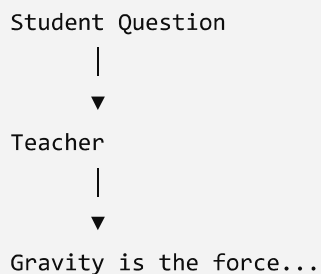
Example 2: Teacher in a Classroom ★

Decoder-Only

A student asks:

"What is gravity?"

The teacher immediately answers.



The teacher generates the response as they speak.

Encoder-Decoder

Now imagine the student hands over a **10-page essay** and says:

"Can you summarize this?"

The teacher first reads the entire essay.

Essay

|

▼

Teacher Reads Everything

|

▼

Writes Summary



Reading first, writing later.

Example 3: Chef 🍕

Decoder-Only

Customer says:

"Give me a pizza recipe."

The chef immediately starts explaining.

Question

↓

Recipe generated
step by step



Encoder-Decoder

Customer brings an existing recipe.

"Convert this recipe into a vegan version."

The chef first understands the recipe.

Original Recipe

↓



Understand Ingredients



Generate Vegan Recipe

Input is transformed into another form.

Example 4: Translator 🌐

This is the classic Encoder-Decoder example.

Suppose someone says:

"I love machine learning."

A translator **cannot** start translating after hearing only:

I...



They usually need the complete sentence.

Entire Sentence



Understand Meaning



Translate



J'aime l'apprentissage automatique.

That's exactly how Encoder-Decoder models work.

Example 5: Lawyer ⚖️

Decoder-Only

Client asks:

"What is copyright?"

The lawyer immediately explains.

Question



↓

Answer

Encoder-Decoder

Client uploads a **200-page contract** and asks:

"Summarize the important clauses."

The lawyer first reads everything.

Contract



↓

Understand

↓

Generate Summary

Example 6: Doctor

Decoder-Only

Patient asks:

"Why do I have a fever?"

Doctor explains possible causes.

Question



↓

Encoder-Decoder

Patient uploads:

- Blood report
- MRI
- ECG
- Medical history

Doctor first reviews all reports.

Reports



↓

Analyze

↓

Diagnosis Summary

Understanding comes before generation.

Example 7: Coding Assistant

Decoder-Only (GitHub Copilot)

You type:

`</> TypeScript`



```
function add(a, b) {
```

The AI predicts:

`</> TypeScript`



```
  return a + b;  
}
```

It simply predicts the next tokens.

Encoder-Decoder

Suppose you ask:

"Convert this Java code to Python."

Java Code



↓

Understand Structure

↓

Generate Python Code

This is a transformation task.

One Analogy That Makes It Easy to Remember

Imagine two professionals:

 **Conversation Expert (Decoder-Only)**

You ask a question.

They reply naturally, one sentence at a time.

Question



↓

Response

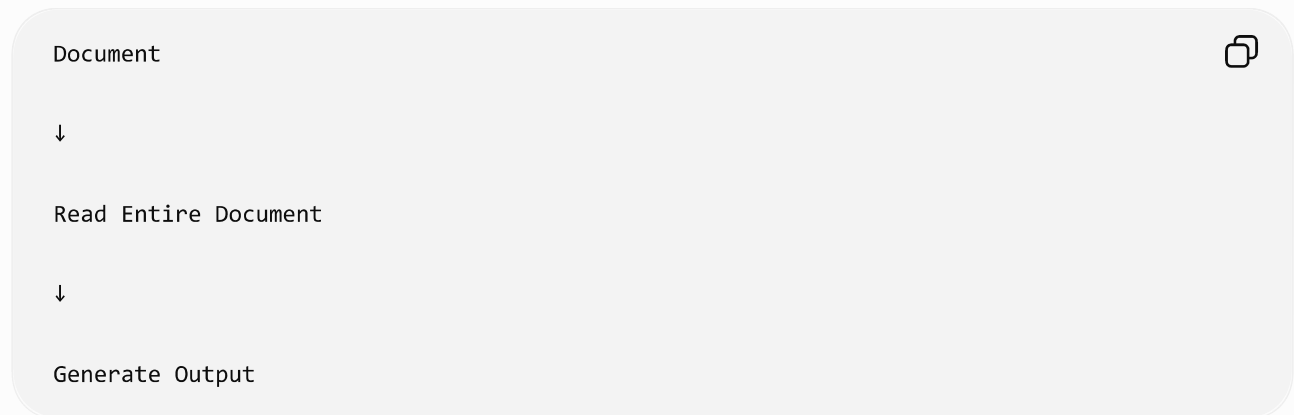
Examples:

- ChatGPT
 - Coding assistant
 - Story writer
-

Document Expert (Encoder-Decoder)

You hand them a document.

They read it completely before responding.



Examples:

- Translator
- Summarizer
- Grammar corrector

Memory Trick for Interviews

Real-Life Person	AI Architecture	Why
Friend answering your question	Decoder-Only	Generates a response as the conversation progresses
Translator	Encoder-Decoder	Understands the complete input before producing the output
Teacher summarizing notes	Encoder-Decoder	Reads everything first, then summarizes
Lawyer reviewing a contract	Encoder-Decoder	Analyzes the full document before responding
Chatbot	Decoder-Only	Produces the next token continuously
Story writer	Decoder-Only	Continues generating text from the existing context

30-Second Interview Explanation

Think of a **decoder-only model** like a person having a conversation. They listen to your question and start responding immediately, generating the answer one word at a time. That's why models like GPT and Llama are excellent for chatbots, coding assistants, and story generation.

An **encoder-decoder model** is like a translator or someone summarizing a document. They first read and understand the entire input, then generate the output based on that understanding. This makes them ideal for tasks such as translation, summarization, and text transformation.



Do you like this personality?

